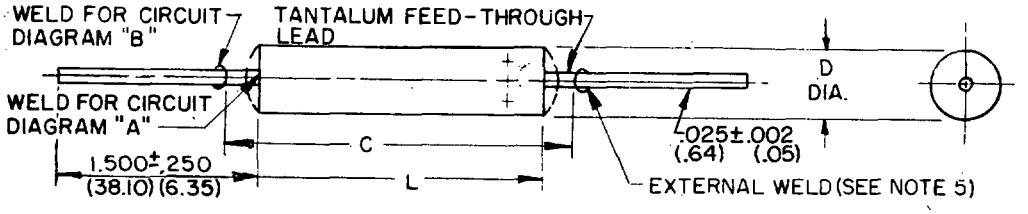
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, PLAIN FOIL) -85°C (VOLTAGE DERATED TO 125°C), STYLES CL30 AND CL31

INACTIVE FOR NEW DESIGN; USE MIL-C-39006/3

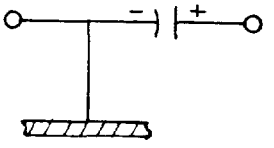
The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-3965.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

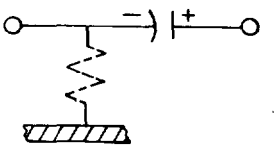


Case Size	Dimensions			
	1/ L ±.062 (1.57)	D ±.016 (.41)	D Max	C Max
		CL30	CL31	
E1	.688 (17.48)	.188 (4.78)	.219 (5.56)	1.188 (30.18)
G2	.969 (24.61)	.281 (7.14)	.312 (7.92)	1.469 (37.31)
G3	1.438 (36.53)	.375 (9.53)	.406 (10.31)	1.938 (49.23)
G4	2.125 (53.98)	.375 (9.53)	.406 (10.31)	2.625 (66.68)
G5	2.750 (69.85)	.375 (9.53)	.406 (10.31)	3.250 (82.55)

1/ For style CL31, nonshrinkable sleeving shall extend .016 inch minimum, .062 inch maximum, beyond each end of the case; shrinkable sleeving shall lap over the ends of the case.



CASE
CIRCUIT DIAGRAM "A"



CASE
CIRCUIT DIAGRAM "B"
(See Note 4)

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4mm.
4. There is an indeterminate resistance between the metal case and the negative terminal.
5. The weld shall not be enclosed in the end seal.

FIGURE 1. Styles CL30 and CL31 capacitors

FSC 5910

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TABLE I. Capacitor characteristics.

Type designation[1/]	Voltage			Capacitance	DC leakage (max)		Impedance (max)	Case size
	Rated (85°C)	Derated (125°C)	Surge (85°C)		25°C	85°C & 125°C		
	volts, dc	volts, dc	volts, dc	µF	µA	µA	ohms	
CL3-BA100MPE	3	2	3.4	10	1	2	250	E1
CL3-BA500MPG	3	2	3.4	50	2	6	50	G2
CL3-BB080MPE	6	4	6.9	8	1	2	329	E1
CL3-BB300MPG	6	4	6.9	30	2	6	88	G2
CL3-BB101MPG	6	4	6.9	100	2	10	26	G3
CL3-BB201MPG	6	4	6.9	200	4	30	13	G4
CL3-BB301MPG	6	4	6.9	300	5	50	9	G5
CL3-BD060MPE	10	7	11.5	6	1	2	440	E1
CL3-BD250MPG	10	7	11.5	25	2	6	105	G2
CL3-BD800MPG	10	7	11.5	80	2	10	33	G3
CL3-BD161MPG	10	7	11.5	160	4	20	17	G4
CL3-BD221MPG	10	7	11.5	220	5	50	12	G5
CL3-BE4R5MPE	15	10	17.2	4.5	1	2	495	E1
CL3-BE100MPG	15	10	17.2	10	2	4	250	G2
CL3-BE180MPG	15	10	17.2	18	2	4	125	G2
CL3-BE550MPG	15	10	17.2	55	2	10	41	G3
CL3-BE111MPG	15	10	17.2	110	4	20	20	G4
CL3-BE161MPG	15	10	17.2	160	5	30	14	G5
CL3-BG030MPE	25	15	28.8	3	1	2	744	E1
CL3-BG120MPG	25	15	28.8	12	2	4	186	G2
CL3-BG350MPG	25	15	28.8	35	2	10	64	G3
CL3-BG700MPG	25	15	28.8	70	4	20	32	G4
CL3-BG101MPG	25	15	28.8	100	6	30	23	G5
CL3-BH2R5MPE	30	20	34.5	2.5	2	5	893	E1
CL3-BH100MPG	30	20	34.5	10	2	6	265	G2
CL3-BH300MPG	30	20	34.5	30	2	13	75	G3
CL3-BH600MPG	30	20	34.5	60	4	26	37	G4
CL3-BH850MPG	30	20	34.5	85	6	38	27	G5
CL3-BJ1R5MPE	50	30	57.5	1.5	1	4	1487	E1
CL3-BJ060MPG	50	30	57.5	6	2	6	372	G2
CL3-BJ200MPG	50	30	57.5	20	2	13	112	G3
CL3-BJ400MPG	50	30	57.5	40	4	26	56	G4
CL3-BJ550MPG	50	30	57.5	55	6	38	41	G5
CL3-BL010MPE	75	50	86.2	1	1	4	2232	E1
CL3-BL040MPG	75	50	86.2	4	2	6	558	G2
CL3-BL140MPG	75	50	86.2	14	2	13	159	G3
CL3-BL280MPG	75	50	86.2	28	4	26	80	G4
CL3-BL400MPG	75	50	86.2	40	6	38	56	G5
CL3-BN0R8MPE	100	65	115.0	0.8	1	3.5	2790	E1
CL3-BN030MPG	100	65	115.0	3	2	6	743	G2
CL3-BN100MPG	100	65	115.0	10	2	13	223	G3
CL3-BN200MPG	100	65	115.0	20	3	25	112	G4
CL3-BN300MPG	100	65	115.0	30	6	38	74	G5
CL3-BQ0R5MPE	150	100	172.0	0.5	2	4	4454	E1
CL3-BQ020MPG	150	100	172.0	2	2	6	1115	G2
CL3-BQ040MPG	150	100	172.0	4	3	13	558	G3
CL3-BQ070MPG	150	100	172.0	7	3	13	319	G3
CL3-BQ080MPG	150	100	172.0	8	4	26	265	G4
CL3-BQ120MPG	150	100	172.0	12	4	26	208	G4
CL3-BQ140MPG	150	100	172.0	14	4	26	159	G4
CL3-BQ200MPG	150	100	172.0	20	6	38	112	G5

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TABLE I. Capacitor characteristics. - Continued

Type designation 1/	Voltage			Capacitance	DC leakage (max)		Impedance (max)	Case size
	Rated (85°C)	Derated (125°C)	Surge (85°C)		25°C	85°C & 125°C		
	volts, dc	volts, dc	volts, dc	µF	µA	µA	ohms	
CL3-BZR35LPE	200	150	230.0	0.35	4	25	5577	E1
CL3-BZ1R5LPG	200	150	230.0	1.5	8	32	1300	G2
CL3-BZ050LPG	200	150	230.0	5	16	64	400	G3
CL3-BZ100LPG	200	150	230.0	10	28	100	200	G4
CL3-BZ150LPG	200	150	230.0	15	40	160	130	G5
CL3-BOR30LPE	250	165	287.0	.30	6	32	6600	E1
CL3-BO1R2LPG	250	165	287.0	1.2	8	40	1600	G2
CL3-BO040LPG	250	165	287.0	4	16	80	500	G3
CL3-BO080LPG	250	165	287.0	8	28	140	250	G4
CL3-BO120LPG	250	165	287.0	12	40	200	160	G5
CL3-BXR27LPE	300	200	345.0	.27	6	40	9000	E1
CL3-BX010LPG	300	200	345.0	1	10	50	2250	G2
CL3-BX030LPG	300	200	345.0	3	20	100	750	G3
CL3-BX070LPG	300	200	345.0	7	35	175	322	G4
CL3-BX100LPG	300	200	345.0	10	50	250	225	G5
CL3-BYR22LPE	375	250	431.0	.22	6	80	11800	E1
CL3-BYR82LPG	375	250	431.0	.82	10	100	2810	G2
CL3-BY2R7LPG	375	250	431.0	2.7	20	225	900	G3
CL3-BY5R6LPG	375	250	431.0	5.6	35	250	450	G4
CL3-BY8R2LPG	375	250	431.0	8.2	50	325	281	G5
CL3-BUR15LPE	450	300	518.0	.15	8	120	21400	E1
CL3-BUR56LPG	450	300	518.0	.56	10	150	5350	G2
CL3-BU2R2LPG	450	300	518.0	2.2	20	400	1600	G3
CL3-BU3R9LPG	450	300	518.0	3.9	35	800	800	G4
CL3-BU6R8LPG	450	300	518.0	6.8	50	1000	535	G5

1/ Complete type designation will include an additional symbol to indicate style.

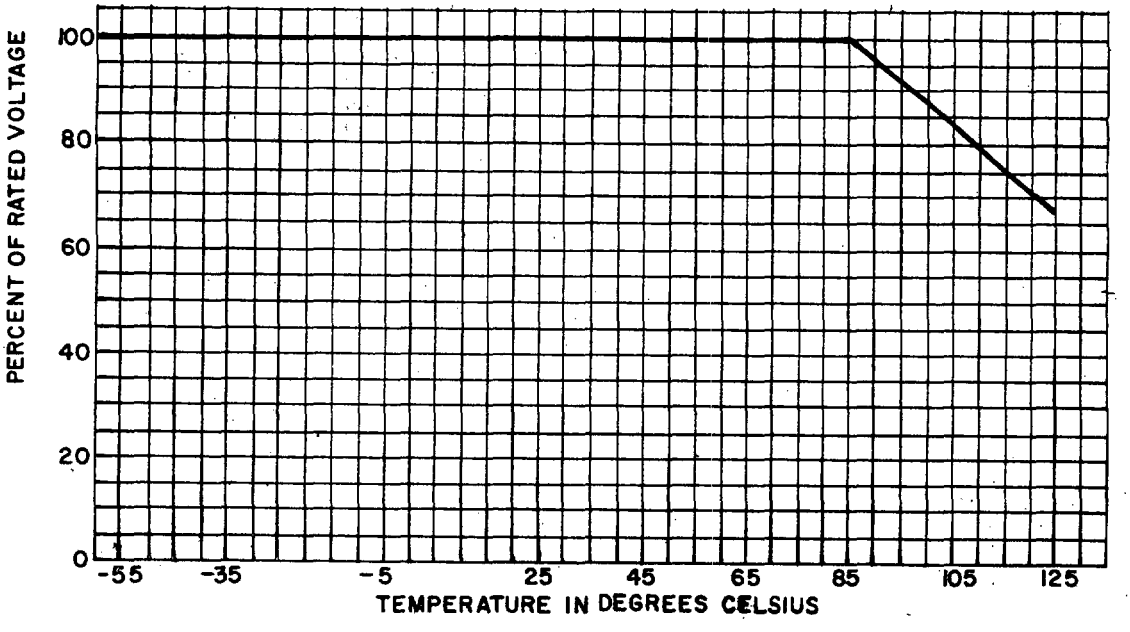


FIGURE 2. Voltage derating with temperature.

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TABLE II SUPERSESSION & CROSS REFERENCE DATA

SUPERSEDED: FROM Type Designation MIL-C-3965/15C	SUPERSEDED: TO Type Designation MIL-C-3965/15D	Cross Reference For New Design Part Number MIL-C-39006/3C
CL3-BA100MPE	CL3-BA100MPE	No Replacement
CL3-BA500MPE	CL3-BA500MPG	No Replacement
CL3-BB080MPE	CL3-BB080MPE	No Replacement
CL3-BB300MPE	CL3-BB300MPG	No Replacement
CL3-BB101MPE	CL3-BB101MPG	No Replacement
CL3-BB201MPE	CL3-BB201MPG	No Replacement
CL3-BB301MPE	CL3-BB301MPG	No Replacement
CL3-BD060MPE	CL3-BD060MPE	No Replacement
CL3-BD250MPE	CL3-BD250MPG	No Replacement
CL3-BD800MPE	CL3-BD800MPG	No Replacement
CL3-BD161MPE	CL3-BD161MPG	No Replacement
CL3-BD221MPE	CL3-BD221MPG	No Replacement
CL3-BE4R5MPE	CL3-BE4R5MPE	M39006/03-2306
CL3-BE100MPE	CL3-BE100MPG	No Replacement
CL3-BE180MPE	CL3-BE180MPG	M39006/03-1349
CL3-BE550MPE	CL3-BE550MPG	M39006/03-1350
CL3-BE111MPE	CL3-BE111MPG	M39006/03-1351
CL3-BE161MPE	CL3-BE161MPG	M39006/03-1352
CL3-BG030MPE	CL3-BG030MPE	M39006/03-2365
CL3-BG120MPE	CL3-BG120MPG	M39006/03-1353
CL3-BG350MPE	CL3-BG350MPG	M39006/03-1354
CL3-BG700MPE	CL3-BG700MPG	M39006/03-1355
CL3-BG101MPE	CL3-BG101MPG	M39006/03-1356
CL3-BH2R5MPE	CL3-BH2R5MPE	M39006/03-2370
CL3-BH100MPE	CL3-BH100MPG	M39006/03-1357
CL3-BH300MPE	CL3-BH300MPG	M39006/03-1358
CL3-BH600MPE	CL3-BH600MPG	M39006/03-1359
CL3-BH850MPE	CL3-BH850MPG	M39006/03-1360
CL3-BJ1R5MPE	CL3-BJ1R5MPE	M39006/03-2375
CL3-BJ060MPE	CL3-BJ060MPG	M39006/03-1362
CL3-BJ200MPE	CL3-BJ200MPG	M39006/03-1363
CL3-BJ400MPE	CL3-BJ400MPG	M39006/03-1364
CL3-BJ550MPE	CL3-BJ550MPG	M39006/03-1365
CL3-BL010MPE	CL3-BL010MPE	M39006/03-2380
CL3-BL040MPE	CL3-BL040MPG	M39006/03-1366
CL3-BL140MPE	CL3-BL140MPG	M39006/03-1367
CL3-BL280MPE	CL3-BL280MPG	M39006/03-1368
CL3-BL400MPE	CL3-BL400MPG	M39006/03-1369
CL3-BN0R8MPE	CL3-BN0R8MPE	M39006/03-2385
CL3-BN030MPE	CL3-BN030MPG	M39006/03-1370
CL3-BN100MPE	CL3-BN100MPG	M39006/03-1371
CL3-BN200MPE1	CL3-BN200MPG	M39006/03-1372
CL3-BN300MPE	CL3-BN300MPG	M39006/03-1373
CL3-BQ0R5MPE	CL3-BQ0R5MPE	M39006/03-2390
CL3-BQ020MPE	CL3-BQ020MPG	M39006/03-1374
CL3-BQ040MPE	CL3-BQ040MPG	No Replacement
CL3-BQ070MPE	CL3-BQ070MPG	M39006/03-1375
CL3-BQ080MPE	CL3-BQ080MPG	No Replacement
CL3-BQ120MPE	CL3-BQ120MPG	No Replacement
CL3-BQ140MPE	CL3-BQ140MPG	M39006/03-1376
CL3-BQ200MPE	CL3-BQ200MPG	M39006/03-1377
CL3-BZR35MPE	CL3-BZR35LPE	M39006/03-2395
CL3-BZ1R5MPE	CL3-BZ1R5LPG	M39006/03-1378
CL3-BZ050MPE	CL3-BZ050LPG	M39006/03-1379
CL3-BZ100MPE	CL3-BZ100LPG	M39006/03-1380
CL3-BZ150MPE	CL3-BZ150LPG	M39006/03-1381
CL3-BOR30MPE	CL3-BOR30LPE	M39006/03-2400
CL3-BO1R2MPE	CL3-BO1R2LPG	M39006/03-1382
CL3-BO040MPE	CL3-BO040LPG	M39006/03-1383
CL3-BO080MPE	CL3-BO080LPG	M39006/03-1384
CL3-BO120MPE	CL3-BO120LPG	M39006/03-1385
CL3-BXR27MPE	CL3-BXR27LPE	M39006/03-2405

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TABLE II SUPERSESSION & CROSS REFERENCE DATA (Cont'd)

SUPERSEDED		
FROM Type Designation[1/] MIL-C-3965/15C	TO Type Designation MIL-C-3965/15D	Cross Reference For New Design Part Number MIL-C-39006/3C
CL3-BX010MPE	CL3-BX010LPG	M39006/03-1386
CL3-BX030MPE	CL3-BX030LPG	M39006/03-1387
CL3-BX070LPE	CL3-BX070LPG	M39006/03-1388
CL3-BX100LPE	CL3-BX100LPG	M39006/03-1389
CL3-BYR22LPE	CL3-BYR22LPE	M39006/03-2410
CL3-BYR82LPE	CL3-BYR82LPG	M39006/03-1390
CL3-BY2R7LPE	CL3-BY2R7LPG	M39006/03-1391
CL3-BY5R6LPE	CL3-BY5R6LPG	M39006/03-1392
CL3-BY8R2LPE	CL3-BY8R2LPG	M39006/03-1393
CL3-BUR15LPE	CL3-BUR15LPE	M39006/03-2415
CL3-BUR56LPE	CL3-BUR56LPG	M39006/03-1394
CL3-BU2R2LPE	CL3-BU2R2LPG	M39006/03-1395
CL3-BU3R9LPE	CL3-BU3R9LPG	M39006/03-1396
CL3-BU6R8LPE	CL3-BU6R8LPG	M39006/03-1397

1/ Complete Type Designation will include an additional symbol to indicate style.

2/ Part Number referenced will replace Styles CL30 and CL31.

REQUIREMENTS:

Design and construction:
Dimensions & configuration-See Fig 1.
Case type-Tubular; uninsulated (CL30), insulated (CL31).
Seal type - Hermetic (symbol G).
- Nonhermetic (symbol E).
Terminals - Axial wire-lead.
Rated temperature- -55°C to +85°C, voltage derated to +125°C (See Fig 2).
Rated voltage - See Table I.
Capacitance (cap)(nom) - See Table I.
Capacitance tolerance-Symbol L (±15%) or M (±20%) as shown in Table I.

DC leakage (DCL)(max)(at +25°C):
See Table I.

Dissipation factor (DF)(max):
15% (≤10v) or 10% (>10v).

Seal: In accordance with MIL-C-3965.

Dielectric withstanding voltage (DWV):
Sleeving to case(min) - 2,000 vdc.
Terminal to case - Not applicable.

Insulation resistance:

Sleeving to case(min): 100 megohms.
Terminal to case - Not applicable.

Temperature cycling:
DCL(max) - 125% of 25°C value of Table I.
Δ Cap-Within ±5% of initial measurement.
DF(max)-115% of initial requirement.

Solderability: In accordance with MIL-C-3965.

Terminal strength:
Pull test-In accordance with MIL-C-3965.
Bend Test - Not applicable.
Torque test - Not applicable.
Wire-lead bend test - In accordance with MIL-C-3965.
Twist test - Not applicable.

Surge voltage: See Table I.

Moisture resistance:
DCL(max) - 125% of 25°C value of Table I.
Δ Cap - Within ±3% of initial measurement.
DF(max)-115% of initial requirement.

Low temperature (storage):
DCL(max)-See table I (25°C).
ΔCap-Within ±3% of initial measurement.
DF(max)-Shall not exceed initial requirement.

Stability at low and high temperature:
Step 1 (+25°C):
DCL (max) - See Table I.
Δ Cap - Within tolerance of Table I value.
DF (max) - Shall not exceed initial requirement.
Step 2 (-55°C):
Impedance - See Table I.
Δ Cap (max) - -25%(≤250v) or -40% (>250v) of step 1 value.
Step 3 (+25°C):
DCL (max) - See Table I.
Δ Cap - Within ±3% of step 1 value.
DF (max) - Shall not exceed initial requirement.

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Step 4 (+85°C):
DCL (max) - See Table I.
ΔCap (max) - +20% (≤50v) or +15% (>50v) of Step 1 value.
DF (max) - 20% (≤50v) or 15% (>50v).
1/ Step 5 (+125°C):
DCL (max) - See Table I.
ΔCap (max) - +25% (≤30v) or +15% (>30v) of step 1 value.
DF (max) - 25% (≤30v) or 15% (>30v).
Step 6 (+25°C):
DCL (max) - See Table I.
ΔCap - Within ±3% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Reverse voltage (>6v):
DCL (max) - 125% of 25°C value of Table I.
ΔCap - Within ±10% of initial measurement.
DF - Shall not exceed initial requirement.

Life at +85°C:
2,000-hr (qualification):
DCL (max) - See table I (25°C).
ΔCap - Within ±15% (≤50v) or ±10% (>50v) of initial measurement.
DF (max) - Shall not exceed initial requirement.

240-hr (Group B):
DCL (max) - See Table I (25°C).
ΔCap - Within ±10% of initial measurement.
DF (max) - Shall not exceed initial requirement.

1,760-hr (group C):
DCL (max) - See Table I (25°C).
ΔCap - Within ±15% (≤50v) or ±10% (>50v) of initial measurement.
DF (max) - Shall not exceed initial requirement.

Barometric pressure:
Test voltage - Rated voltage.

1/ Life at +125°C:
DCL (max) - See Table I (25°C).
ΔCap - Within ±15% (≤30v) or +10% (>30v) of initial measurement.
DF (max) - Shall not exceed initial requirement.
DWV (min)(CL31only) - 2000 vdc.
IR (min)(CL31 only) - 100 megohms.

Marking:
Polarity - Series of plus (+) symbols encircling positive end (see Figure 1).

Voltage groups (85°C rated voltage):
I -------- 15 to 50v, incl.
II -------- 75 to 200v, incl.
III -------- 250 to 450v, incl.

1/ 125°C derated voltage referenced.

NOTE: The capacitors covered by this specification sheet are substitutable for those covered by MIL-C-3965/15C. See Table II.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review Activities:
Army - EL, MU
Navy - EC, SH
Air Force - 11, 17
DSA - ES

User Activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing Activity:
Air Force - 80

(Project 5910-1048-1)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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